Gezhi: A Local-First AI Study Companion Built on Learning by Teaching

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Abstract

Two strands of evidence define the problem this project addresses. A randomised controlled trial with roughly 1,000 secondary students found that unrestricted access to a large language model improved practice performance by 48% but *reduced* examination performance by 17%; a carefully engineered hint-only tutor produced no examination benefit at all. A separate two-year cluster randomised trial of an AI tutor across 18 middle schools found effects of only 0.06–0.08 standard deviations, comparable to the same platform without AI, because the median student used the tutor on just one third of practice days. Together these results suggest that AI tutoring currently either harms learning or goes unused, and that neither making the model more capable nor making it more careful addresses the underlying cause.

Gezhi is a Windows desktop study companion for secondary-school students in Macau that attacks both failure modes at once. It requires the student to *produce* an explanation rather than consume one, and it delivers that requirement proactively through a scheduling system rather than waiting to be opened. The student teaches an on-screen companion, Xiaoge, who is deliberately positioned a year below them and who holds a fixed library of misconceptions drawn from the student's own error statistics. The companion argues back; it concedes only when the explanation genuinely addresses the misconception. Explanations are scored on a six-dimension rubric, and the resulting mastery estimate schedules both questions and future explanations. Retrieval runs locally over the student's own textbook using a trigram full-text index with page-level citations.

Keywords: AI in education; learning by teaching; teachable agents; retrieval practice; metacognitive calibration; local-first software; SDG 4.

1. Background

1.1 The promise and the first evidence

Generative AI has been widely promoted as the technology that will finally deliver one-to-one tutoring to every learner. The evidence accumulating since 2024 is considerably more complicated, and it is worth stating precisely, because it shaped every design decision in this project.

Bastani et al. (2024) conducted a randomised controlled trial with approximately 1,000 high-school students in Turkey learning mathematics, assigning them to three conditions: business-as-usual practice, practice with unrestricted access to a general-purpose chatbot, and practice with a version of the same model engineered to provide hints without revealing answers. Students in the unrestricted condition solved 48% more practice problems correctly — and then scored **17% worse** on the examination once the AI was removed. Students in the hint-only condition improved their practice performance by 127%, but their examination performance was statistically indistinguishable from the control group that used no AI at all. The authors found that the model's step-by-step solutions were wrong 42% of the time, and that students overwhelmingly used it as a source of answers rather than as a tutor.

A second study addresses a different failure mode. Oreopoulos and Low (2026) report a two-year cluster randomised trial of Khanmigo, an AI tutor deliberately configured to coach rather than answer, across 18 middle schools. The intent-to-treat effect was 0.06–0.08 standard deviations over a school year, with an implied effect of 0.14 SD for a full year of active participation — a range comparable to the same platform used *without* AI assistance. The authors are explicit about the cause, and it

is behavioural rather than technical: 96% of students tried the tutor at least once, but the median student messaged it on only one third of the days on which they practised, and in only **17% of the sessions in which they made a mistake**. Their conclusion is that "realizing the promise of AI tutoring will require getting students to use it, not just giving them access."

1.2 What the two results jointly imply

Taken together, these findings carve out a narrow space in which an AI learning tool can plausibly help:

1. **Making the model answer better is not the solution.** The unrestricted condition answered most practice questions correctly and still harmed examination performance.
2. **Making the model more restrained is also not sufficient.** The hint-only tutor was well designed and produced no examination gain.
3. **The binding constraint is who produces the content.** The only condition that has never been shown to harm learning is the one in which the learner generates the explanation.
4. **Delivery is as important as pedagogy.** An intervention that must be voluntarily opened will, empirically, not be opened.

Points 3 and 4 are the design brief for Gezhi. The first is addressed by making explanation, not consumption, the central act of the application. The second is addressed by making explanations schedulable items in a spaced-repetition queue that surfaces on the application's first screen.

1.3 The local context

Macau secondary students work from a mixture of Chinese-language and English-language materials, frequently in scanned form, and face a high-stakes examination structure. Two practical consequences shaped the system: full-text retrieval must work correctly on Chinese text without a server, and scanned material must be ingestible. Both are addressed in Section 5.2.

1.4 Contribution

This report describes (i) the design of a teachable-agent study companion for secondary learners, in which deliberate and *consistent* misconception is a first-class system component; (ii) a local-first retrieval architecture that grounds the companion's challenges in the student's own textbook; (iii) the coupling of explanation quality to a spaced-repetition schedule; and (iv) an evaluation protocol and the results obtained with secondary-school participants in Macau.

2. Problem Definition

2.1 Research questions

RQ1 — Explanation quality.
Can a teachable companion that is engineered to hold consistent misconceptions, and that refuses to concede prematurely, improve the quality of a secondary student's spoken explanation of a knowledge point, as measured by a six-dimension rubric?

RQ2 — Grounded challenge.
Can a fully local retrieval layer over the student's own textbook supply the companion with challenges that are specific enough to be useful and citable enough to be verifiable, without sending any content to a server?

RQ3 — Proactive delivery.
Does scheduling *explanations* — not only questions — in the review queue increase the frequency with which students actually explain, relative to an application in which explaining must be initiated manually?

RQ4 — Calibration.
Does repeated teach-and-defend practice reduce the gap between a student's self-assessed understanding and their measured explanation quality?

2.2 Target users and scope

The primary user is a secondary-school student in Macau, aged approximately 14–18, studying towards school examinations and university entrance. The design target is specifically the student who believes they understand a topic but cannot articulate it — a misjudgement that is well documented. Across 115 studies, 15,889 participants and 502 effect sizes, the weighted mean correlation between learners' judgements of their own learning and their actual performance is only $r = .178$ (Yang et al., 2022). The system is not designed for primary-school children, nor for learners with no prior exposure to the material: several of the techniques it relies upon require a knowledge base to operate on (Section 3.3).

2.3 Non-goals

Gezhi is not a homework-answering tool, does not generate answers to be copied, and does not attempt to replace classroom instruction. It contains no social or competitive features, by design: the application is single-user and local, and no student data is compared across users or transmitted off the device.

3. Related Work

3.1 Learning by teaching and teachable agents

The pedagogical value of preparing to teach is long established. Chase, Chin, Oppezzo and Schwartz (2009) demonstrated a *protégé effect*: learners who believed they would teach a virtual student invested more effort and achieved better outcomes than learners who studied for themselves. That work, and the broader teachable-agent literature, established the mechanism Gezhi relies on — responsibility for another's understanding changes how a learner processes material.

3.2 LLM-based teachable agents, 2024–2026

Recent work has moved this idea onto large language models, and has produced two findings that directly shaped our design.

Yang, Pujara and Li (2025) inverted the standard tutor relationship, having students act as instructors who must teach a model to solve problems. Their central design device is the *engineered knowledge gap*: questions constructed so that only the student holds the information required to answer them. In an undergraduate course, this produced statistically significant improvement over historical cohorts. The paper notes explicitly that a model which answers too accurately removes the learning opportunity altogether.

Rogers et al. (2025), in work whose title — "Playing Dumb to Get Smart" — states the design principle, created and evaluated an LLM-based teachable agent and found that the agent must present itself as genuinely less capable for the interaction to be pedagogically productive. A related study of a model used as a teachable agent in programming education reported gains in knowledge and self-regulated learning but **no significant improvement in error-correction ability**, for a specific and instructive reason: the model tended to supply correct code directly, so students never got the opportunity to debug anything.

That null result is the single most important piece of evidence for Xiaoge's design. It demonstrates that a teachable agent which is helpful in the ordinary sense is pedagogically inert. The agent must be *reliably and consistently wrong* in specific, diagnosable ways.

3.3 Established techniques, and what remains unaddressed

To avoid rebuilding what already exists, we surveyed the current landscape of AI study tools. The following are effectively commoditised: adaptive spaced-repetition scheduling (with open-source algorithms now outperforming earlier fixed-interval methods on large-scale benchmarks), knowledge-graph-driven question recommendation, Socratic dialogue tutoring, and one-click conversion of source material into flashcards, quizzes or audio. These are entry requirements rather than differentiators.

By contrast, three areas remain largely unaddressed in consumer products: role reversal, in which the student teaches and the AI exposes its own misconceptions; deliberately designed pretesting, in

which the learner attempts a problem before instruction; and proactive, evidence-driven delivery of the learning task at the moment it is needed. GEZHI is positioned in the first and third of these.

We also adopted, with appropriate caution, techniques with strong but bounded evidence. Interleaved practice has a moderate overall effect ($g = 0.42$; Brunmair & Richter, 2019) and, uniquely among the techniques we considered, direct classroom evidence at secondary level: in a three-month study with 126 seventh-grade students, interleaved practice produced $d = 0.42$ on an immediate test and $d = 0.79$ on a test 30 days later (Rohrer, Dedrick & Stershic, 2014). The same meta-analysis reports that interleaving *harmed* vocabulary learning ($g = -0.39$), so GEZHI restricts interleaving to mathematics and science and disables it for language vocabulary.

4. System Design

4.1 Architecture

GEZHI is a single-user Windows desktop application. Its defining architectural property is that the boundary of what the AI may touch is enforced by the operating system rather than by instructions in a prompt.

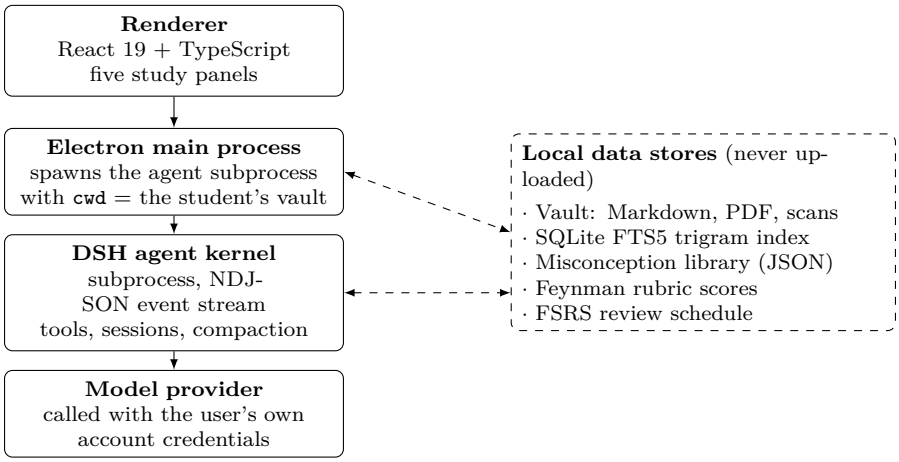


Figure 1: System architecture. The agent kernel runs as a separate process whose working directory is the student's vault; its write boundary is derived from that directory, so it cannot modify anything outside it. All retrieval, scoring and scheduling data remain on the device.

4.2 The learning loop

The application is organised around a single recurring sequence, shown in Table 1. The ordering is deliberate: the student attempts the material before it is explained, commits to a self-assessment before receiving feedback, produces the explanation themselves, and is then challenged on it.

4.3 Xiaoge: engineering a reliably wrong companion

The companion is not an assistant. It is positioned as a student one year below the user, and its function is to make the user's explanation load-bearing.

Three properties are essential and each required explicit engineering:

- **Stable identity.** The companion never evaluates, never says "you are wrong", and never compares the user to anyone.
- **Consistent misconceptions.** The companion's errors come from a structured library (Section 5.5), not from sampling noise. Consistency matters more than the errors themselves: if the companion's objection shifts each turn, the student learns that the system is unreliable rather than that they have a gap to close.
- **Hard refusal to concede prematurely.** Language models are strongly disposed to agree. The system prompt therefore imposes an explicit constraint: the companion may not signal understanding until the student's explanation satisfies the specific correction criterion attached to the active

#	Stage	What happens
0	Trigger	The review queue surfaces a due knowledge point on the first screen.
1	Pretest	The student attempts one item and may answer "I don't know". No feedback is given yet.
2	Self-rating	The student rates their own understanding on a 1–5 scale.
3	Explain	The student explains the topic aloud to XIAOGE. The companion does not interrupt.
4	Challenge	XIAOGE raises one of its stored misconceptions and argues. The student must resolve it.
5	Score	A six-dimension rubric scores the explanation against the textbook source.
6	Repair	The system generates targeted items for the specific gaps identified.
7	Schedule	Both questions and future explanations are scheduled by an FSRS-based model.
8	Delayed self-rating	On a later day, the student re-rates their understanding *after* attempting retrieval.

Table 1: The GEZHI learning loop. Stages 1, 2 and 8 correspond to pretesting, immediate calibration and delayed calibration respectively.

misconception. Without this constraint the interaction degenerates within two turns into dictation to a recording device.

The companion carries a three-by-three lattice above its head, which doubles as a progress display: nine empty cells mean it understands nothing, and each resolved misconception lights one cell in vermilion. This gives the student feedback on progress without presenting a score, which is intended to keep the framing on teaching rather than on being graded.

5. Technical Methods

5.1 Agent kernel: subprocess isolation with an event stream

Rather than implement an agent loop, GEZHI embeds DeepSeek Harness (DSH), an MIT-licensed agent kernel, as a headless subprocess and consumes a newline-delimited JSON event stream. Each event carries a type — session, status, thinking, text, tool call, tool result, final — together with token accounting. This provides session persistence, a tool sandbox, context compaction and an approval policy without requiring a student team to reimplement them, and it keeps the model credential in the user's own account rather than in the application.

The kernel derives its filesystem write boundary from its working directory, so launching it with the working directory set to the student's vault produces a boundary enforced by the process model. Two implementation details proved necessary in practice. First, the subprocess must be started through its command wrapper rather than by invoking the host executable directly, because the installation path contains spaces. Second, the event stream must be decoded as UTF-8 explicitly; reading the redirected byte stream with a non-UTF-8-aware reader corrupts Chinese content, and the corruption occurs on the reading side only. A regression test that writes Chinese text and reads it back is part of the build.

5.2 Local retrieval: a measured tokeniser fix

Retrieval is performed entirely on the device using SQLite, accessed through the runtime's built-in module so that no native dependency is required. We measured behaviour rather than assuming it, and found a failure that would have silently disabled retrieval for the primary user population.

With the default `unicode61` tokeniser, a full-text index returns **zero** results for Chinese substring queries, because the tokeniser treats an entire Chinese sentence as one token. Switching the index to `tokenize='trigram'` resolves this. Measured results over a 200-document corpus: Chinese queries return 0 hits under `unicode61` and 200 hits in approximately 1 ms under `trigram`, at a write cost of roughly 20% and a modest index size increase. Indexing the corpus of 200 documents of 14,400 characters took 61 ms and 73 ms respectively.

Because the tokeniser is fixed at table-creation time, the schema is created once with `trigram` and a migration path exists for databases created with the default. We deliberately did not make vector

search a precondition: keyword retrieval over a trigram index is sufficient for the current corpus size, and adding an embedding dependency would raise the installation cost on a student laptop.

5.3 Citation-forced generation

Every substantive claim the system presents must be traceable to a retrieved passage with a page reference. The retrieval layer returns chunks of 300–500 characters with page metadata, and the generation step is constrained to cite from those chunks. This serves two purposes: it allows the student to verify the companion's challenge against their own textbook, and it makes the companion's misconceptions diagnosable, since a student can see which passage contradicts the objection.

5.4 Explanation scoring

Explanations are scored with a six-dimension rubric, returned as strictly structured JSON so that scores persist and are comparable across sessions (Table 2). Each dimension is scored 1–5. The rubric is applied against the retrieved textbook passage rather than against the model's general knowledge, which keeps scores tied to the material the student is actually being examined on.

Dimension	What it measures
Accuracy	Whether the content of the explanation is correct with respect to the source.
Completeness	Whether all elements the source treats as required are present.
Causality	Whether the student explains *why*, rather than restating *what*.
Plain language	Whether the explanation avoids unexplained jargon.
Example quality	Whether a concrete, correctly chosen example is given.
Gap awareness	Whether the student flags what they are unsure about, rather than glossing over it.

Table 2: The six-dimension explanation rubric. Scores are stored per knowledge point and feed the review scheduler.

5.5 Misconception library

The companion's objections are drawn from a structured library rather than generated freely, which is what makes them consistent and diagnosable. Each entry is stored as:

```
{ misconception_id, surface_claim, why_wrong, correct_point, trigger_condition }
```

`surface_claim` is the plausible but incorrect statement the companion will make; `correct_point` is the criterion the student's explanation must satisfy before the companion concedes; `trigger_condition` determines which knowledge points activate the entry. An illustrative entry for English grammar: the surface claim "*must* and *may* both express possibility, so they can be used interchangeably," with the correction that *must* expresses near-certainty while *may* expresses roughly even likelihood.

Entries are derived from the student's own error statistics in the question bank, so the misconceptions the companion holds are the ones the student actually makes.

5.6 Interleaved practice generation

The examination builder includes an interleaved mode that alternates knowledge points within a single paper and, critically, does not label which topic each item belongs to — labelling removes the discriminative demand that produces the effect. Following the meta-analytic finding that interleaving harms vocabulary acquisition, the mode is enabled only for mathematics and science item types.

5.7 Companion animation and speech

The on-screen companion is implemented as a set of pre-rendered layers that are composed at runtime — body, head, eye states, brow states, five mouth shapes and the head lattice — rather than as frame-by-frame animation. Blinking, breathing and nodding are driven procedurally. Mouth shapes are selected from the speech synthesiser's phoneme-level timestamps, so lip synchronisation requires no audio analysis. The animation format is animated WebP rather than GIF, because GIF's one-bit

transparency produces visible edge artefacts when the character is composited over the application interface.

Two behavioural rules reflect the pedagogical framing. First, expressions lag the content rather than anticipate it: when the student makes an error, the companion first responds with encouragement and only subsequently displays confusion, so that it reads as a peer working through a problem rather than as an examiner. Second, the encouraging state is triggered at least twice as often as the confused state.

5.8 Examination typesetting

The question bank exports typeset examination papers through XeLaTeX, which is required for correct CJK typesetting. Where a system TeX distribution is present it is reused; where it is absent the application falls back to printing through the embedded browser engine, so that a missing TeX installation degrades the output quality rather than breaking the feature.

6. Data Sources

Gezhi ships with no textbook content. The knowledge base is populated by the student from material they already possess, which is both a legal and a pedagogical decision: it removes any question of redistributing copyrighted textbooks, and it ensures the companion's objections concern the material the student is actually assessed on.

Source	Type	Status and use
Student course textbooks and handouts	PDF, Word, Markdown	Supplied by the user; not redistributed.
Scanned worksheets and past papers	Image PDF	Ingested via OCR; OCR is a required path, not an optional one.
User's class notes	Markdown	Created inside the application.
Informatics knowledge-point outlines	Markdown	User-authored; imported as a knowledge-graph skeleton.
Publicly licensed question sets	Text	Imported only where the licence permits; recorded in an asset register.

Table 3: Data sources. All material is user-supplied or explicitly licensed; no copyrighted textbook content is bundled with the application.

Licence discipline. Dependencies are restricted to permissive licences (MIT, BSD, Apache-2.0) suitable for a closed-source distribution. Components under AGPL were identified during selection and excluded from the shipped product. Every external asset is recorded in an asset register with its source URL and licence text.

7. Testing and Evaluation

7.1 Participants

[TO BE COMPLETED: number] secondary-school students from **[TO BE COMPLETED: school name]**, Macau, in Secondary **[TO BE COMPLETED: year levels]**, aged **[TO BE COMPLETED: range]**. Participation was voluntary and required written consent from the student and a parent or guardian. **[TO BE COMPLETED: number]** subject teachers observed or reviewed the sessions.

7.2 Procedure

Each participant completed a single session of approximately **[TO BE COMPLETED: duration]** minutes, followed by a delayed session on the following day:

1. **Pre-test** (5 items on the target knowledge point), followed by a self-rating of understanding on a 1–5 scale.
2. **Teach-and-defend**, in which the participant explained the knowledge point aloud to Xiaoge and responded to its objections until either the misconception was resolved or a limit of **[TO BE**

COMPLETED: number] turns was reached.

3. **Immediate rubric scoring** of the explanation.
4. **Delayed session** (next day): a parallel form of the pre-test, followed by a second self-rating administered *after* the retrieval attempt.
5. **Short questionnaire** on perceived usefulness and perceived difficulty.

7.3 Measures

	Measure	Definition
M1	Explanation quality	Total six-dimension rubric score (6–30) at stages 3 and 6 of the loop.
M2	Immediate calibration gap	Absolute difference between self-rating (scaled to 30) and rubric total, at stage 2.
M3	Delayed calibration gap	The same difference measured after the delayed retrieval attempt (stage 8).
M4	Turns to resolution	Number of companion turns required to resolve the target misconception.
M5	Delayed recall	Score on the parallel test form, administered one day later.

Table 4: Evaluation measures. M2 and M3 follow the finding that delaying a judgement of learning improves its relative accuracy substantially, while leaving overall optimism largely unchanged.

7.4 Analysis

Paired comparisons use Wilcoxon signed-rank tests given the expected sample size, with matched-pairs rank-biserial correlation reported as an effect size. We report M5 as a descriptive outcome rather than as an efficacy claim, and we caution against interpreting any result from a single-arm design as causal (Section 9).

8. Results

[TO BE COMPLETED: This entire section must be completed with the team's own measured data before submission. Do not estimate, round up, or infer any value. If a measurement was not taken, report it as not taken.]

Measure	Pre	Post	Change	Test statistic
M1 Explanation quality	[mean ± SD]	[mean ± SD]	[value]	[statistic, p, effect size]
M2 Calibration gap (immediate)	—	[value]	—	—
M3 Calibration gap (delayed)	—	[value]	[vs M2]	[statistic, p]
M4 Turns to resolution	—	[median (IQR)]	—	—
M5 Delayed recall	[value]	[value]	[value]	[statistic, p]

Table 5: Results. **[TO BE COMPLETED: fill from the session logs exported by the application.]**

8.1 Qualitative observations

[TO BE COMPLETED: Two or three specific, quotable observations from the sessions. Useful examples: a participant who changed their explanation after the third objection; a participant who initially rated their understanding highly and revised it downward after failing the delayed retrieval; any case where the companion's misconception was itself wrong and the student correctly identified that. The last case is particularly valuable and should not be omitted.]

8.2 Deviations from the protocol

[TO BE COMPLETED: Record honestly any session that did not follow the procedure — missing data, a participant who stopped early, a technical failure. Reporting these strengthens rather than weakens the report.]

9. Limitations

We state these plainly, because the evidence base described in Section 1 is precisely about the danger of over-claiming.

- **No control group.** The evaluation is single-arm, so improvements cannot be attributed to Gezhi rather than to practice, novelty, or the attention of being observed. Claims are limited to feasibility and to within-participant change.
- **Small sample, single site.** **[TO BE COMPLETED: n]** participants at one school cannot support generalisation to Macau secondary education.
- **Short duration.** A single session with a one-day delayed test measures immediate retention, not durable learning. The spaced-repetition behaviour that the design depends on would require weeks to evaluate.
- **Self-report instruments.** Self-ratings of understanding are the object of study in M2 and M3 and are simultaneously self-reported, which is a confound we cannot remove within this design.
- **Model non-determinism.** The companion's objections vary between runs even with a fixed misconception library, because the surface wording is generated. This affects reproducibility of the interaction, though not of the underlying misconception.
- **Rubric validity.** The six-dimension rubric is our own construction. It has not been validated against external raters, and inter-rater reliability with human teachers has not yet been established.
- **Novelty is narrower than it may appear.** Teachable agents are an established research area, and 2024–2026 has produced several LLM-based systems of this kind. Our contribution concerns the population, the grounding of the challenge in the student's own material, the coupling to scheduling, and the local-first constraint — not the invention of the mechanism.

10. Ethics, Safety and Privacy

10.1 Data protection

The knowledge base, the learning record and all scores reside on the student's device. No student content is uploaded to a server operated by the team, and no analytics are collected. Model inference is performed through the user's own account, and prompts are not retained by the application. The agent kernel's write boundary is derived from the working directory at process launch, so the AI cannot modify files outside the student's vault; this is enforced by the operating system rather than by instruction.

Where a session requires network access, this is disclosed before the session begins. The application degrades to local retrieval, local speech recognition and a text-only mode rather than failing.

10.2 Research with human participants

Participation was voluntary. Informed consent was obtained in writing from each student and from a parent or guardian, covering the purpose of the study, the data collected, and the right to withdraw at any time without consequence. Participants were assigned codes; no names, student numbers, faces or voices appear in any submitted material. Audio recordings were used only for transcription during the session and were deleted afterwards. The study was conducted with the knowledge and approval of the participating school.

10.3 Responsible AI use and academic integrity

Generative AI tools were used in this project for software development, for the analysis of published research, and for the preparation of design assets. The complete disclosure is in Appendix A. All

reported data were produced by the student team through sessions they ran themselves. No result in this report is generated, estimated or inferred by a model. External research findings are cited to their original sources, and where a widely circulating figure could not be verified against a primary source we have said so rather than repeating it.

10.4 Copyright

No copyrighted textbook content is distributed with the application. The knowledge base is built from material the user supplies. Third-party libraries are restricted to permissive licences, and every external asset is recorded with its source and licence.

10.5 Bias and safety

Two risks were considered explicitly. First, the misconception library is drawn from the student's own errors and therefore cannot introduce misconceptions the student does not already hold; this limits the companion's ability to propagate novel incorrect content. Second, the companion is instructed never to evaluate the student as a person, never to compare them with others, and never to discourage. Encouragement is triggered more frequently than confusion by design, so that the interaction reads as collaborative rather than as assessment.

11. Conclusion and Future Work

Gezhi began from two findings that appear to leave no room for an AI learning tool: an AI that answers questions harms examination performance, and an AI that does not answer questions goes unused. Both failures share a cause — the student is not the one producing the content — and that is the constraint the system is built around. The student explains; an on-screen companion that is deliberately and consistently wrong makes that explanation load-bearing; and the whole interaction is scheduled rather than left to the student's initiative.

The immediate next steps are to complete a controlled comparison against a no-AI condition, to validate the scoring rubric against human raters, and to run the evaluation over a period long enough for the spaced-repetition schedule to matter. Beyond that, the most promising extension is to make the companion's misconceptions visible to teachers as a class-level picture of what students actually misunderstand — which is information that current assessment, focused on right and wrong answers, does not capture.

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A. AI Tool Disclosure

The competition rules require teams to state the generative AI tools used, their versions, their purpose, and the parts completed by the students. This appendix is that statement.

Tool	Version	Purpose	Completed by students
[tool name]	**[version]**	**[purpose]**	**[what the team did themselves]**
[tool name]	**[version]**	**[purpose]**	**[what the team did themselves]**
[tool name]	**[version]**	**[purpose]**	**[what the team did themselves]**

Table 6: Generative AI tools used in this project. **[TO BE COMPLETED: Complete every row before submission; this table is checked against the competition's academic integrity rules.]**

Division of work. The problem identification, the research design, the software architecture, the misconception library, the evaluation sessions and the analysis of results were carried out by the student team. Generative AI was used as a development assistant (code generation and review, documentation) and as a research aid (locating and summarising published literature, all of which was then read at source before citation). No reported data were generated or estimated by an AI system.